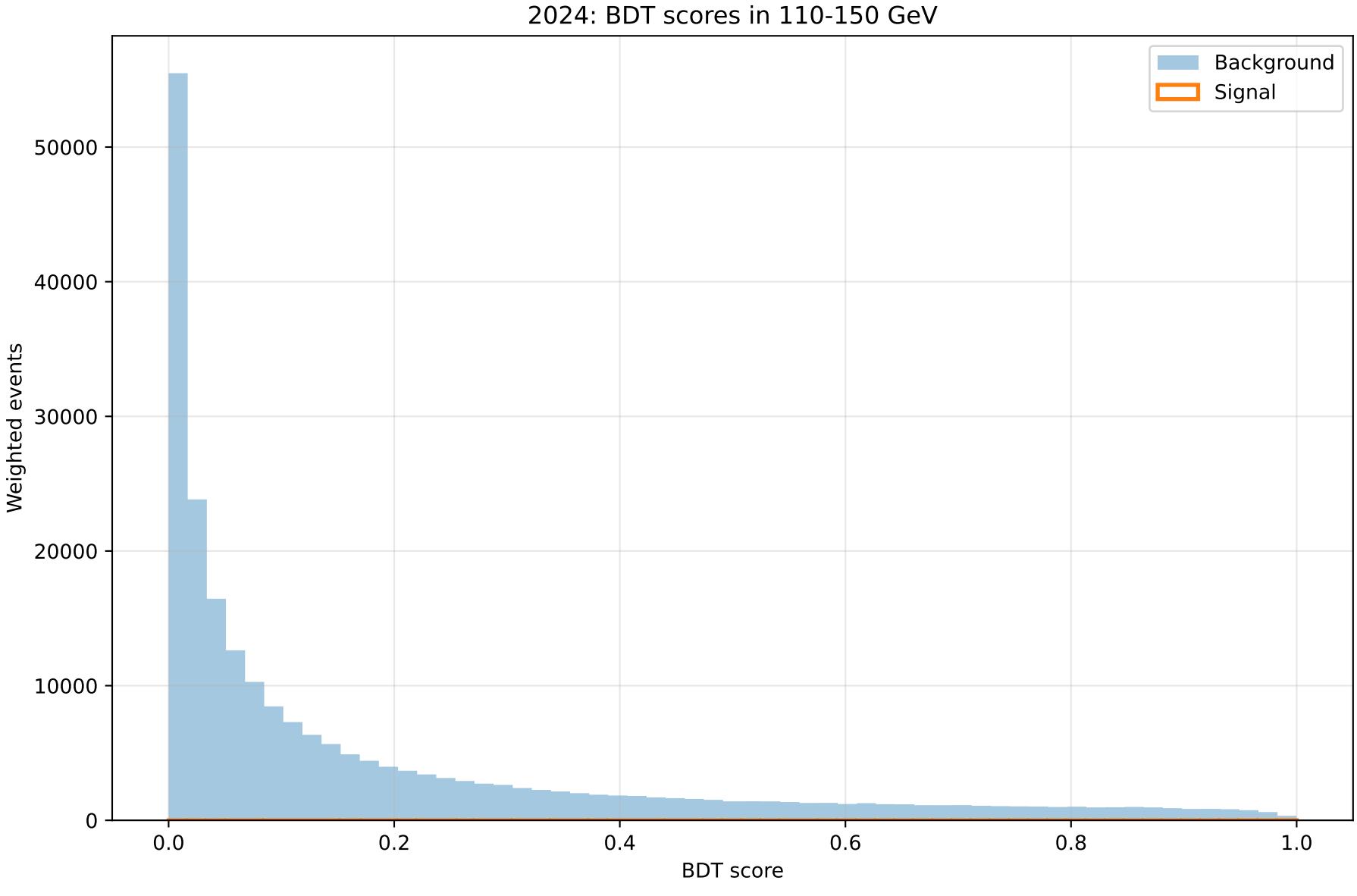
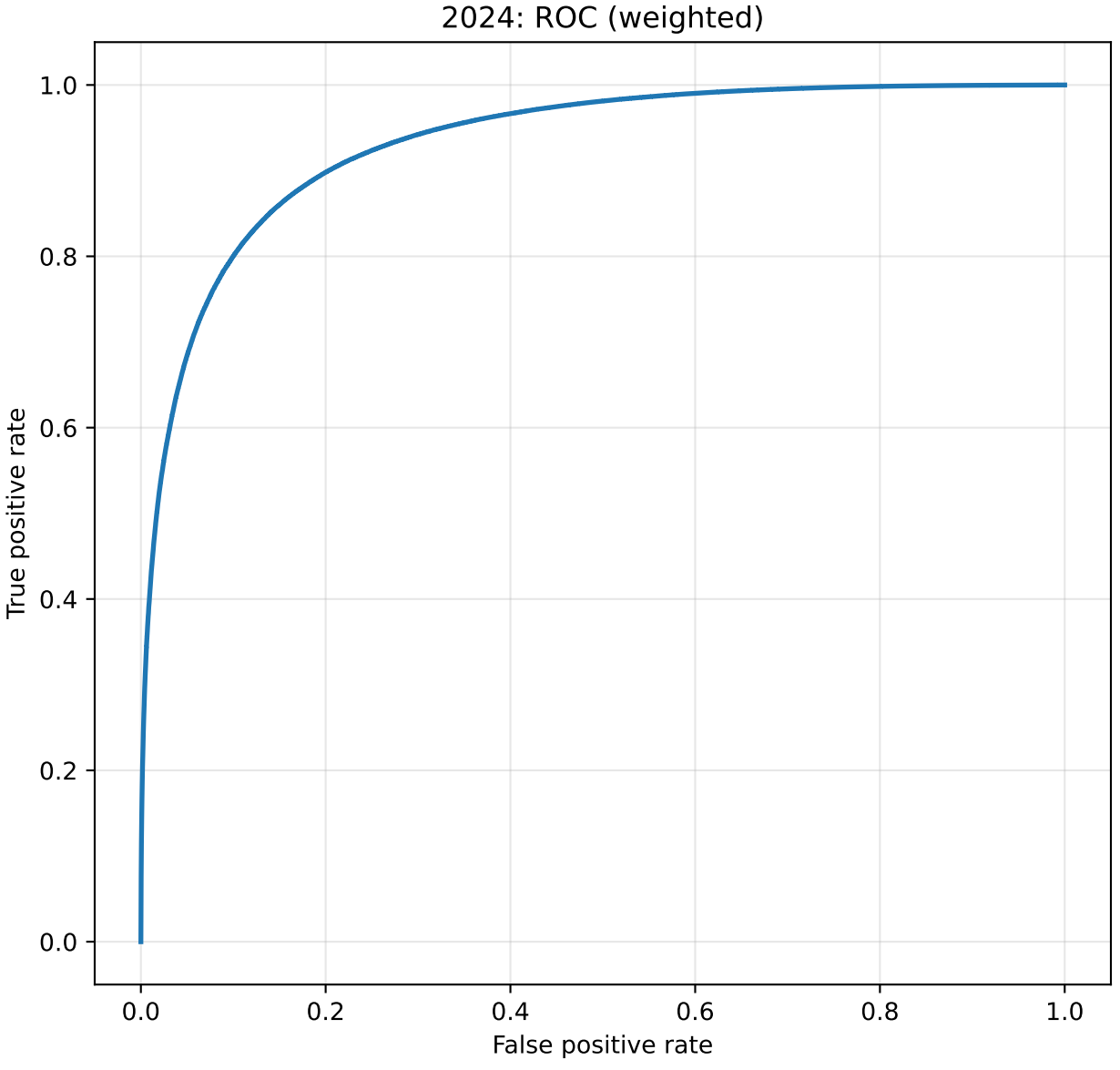
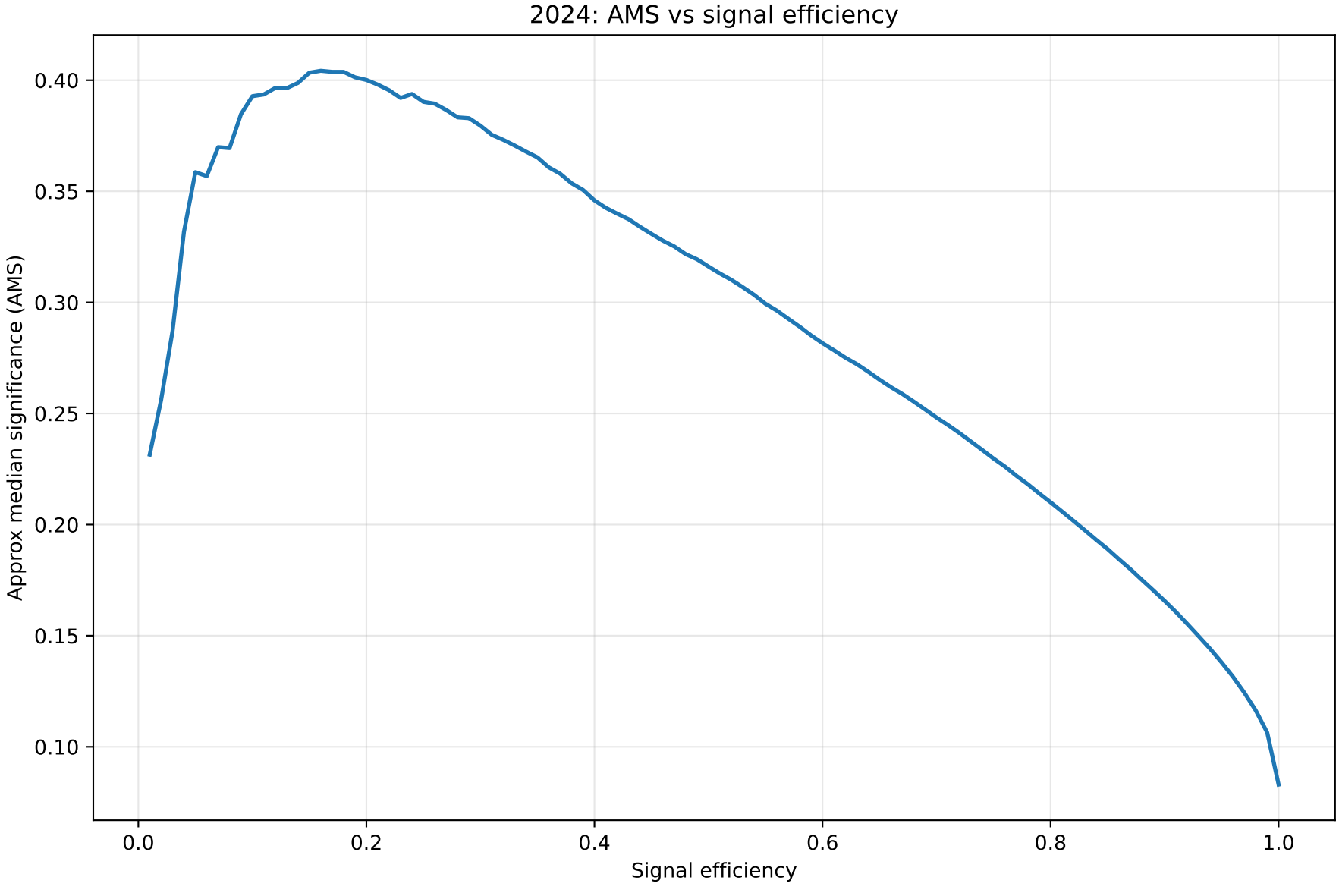
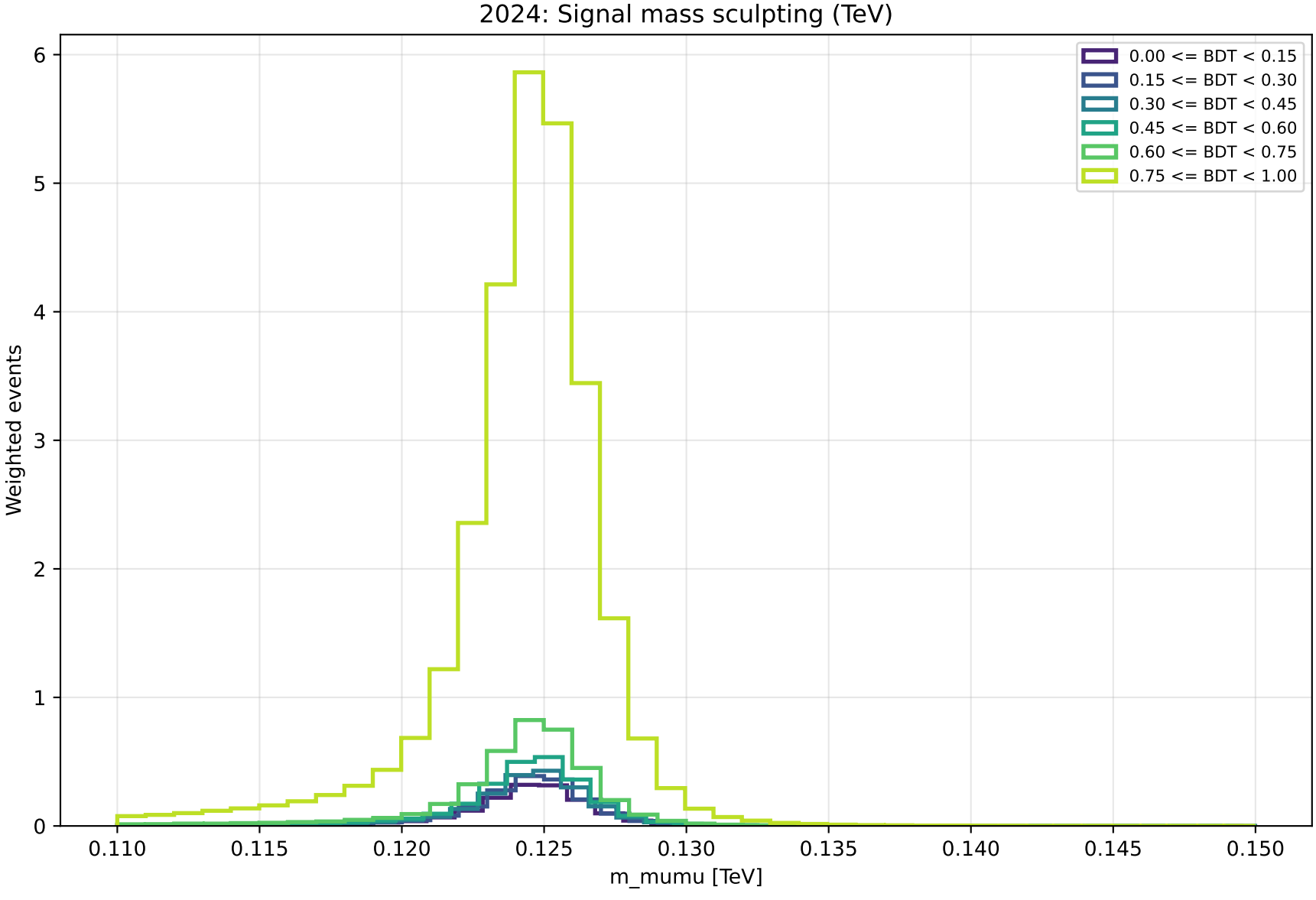

2024: BDT scores in 110-150 GeV
Background
Signal
Weighted events
50000
40000
30000
20000
10000
0
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

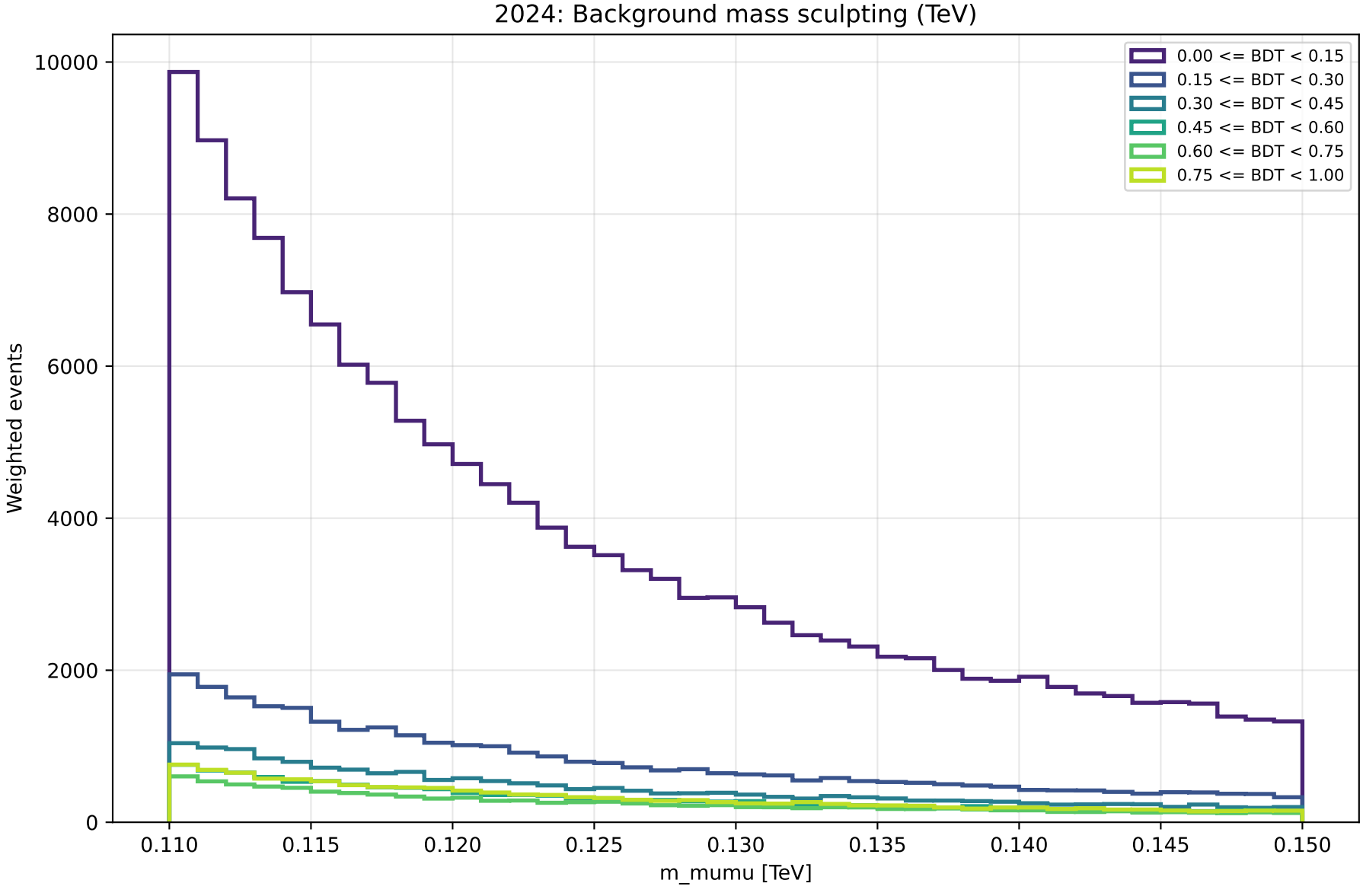

2024: ROC (weighted)
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0


2024: AMS vs signal efficiency
Approx median significance (AMS)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.0
0.2
0.4
0.6
0.8
1.0
Signal efficiency

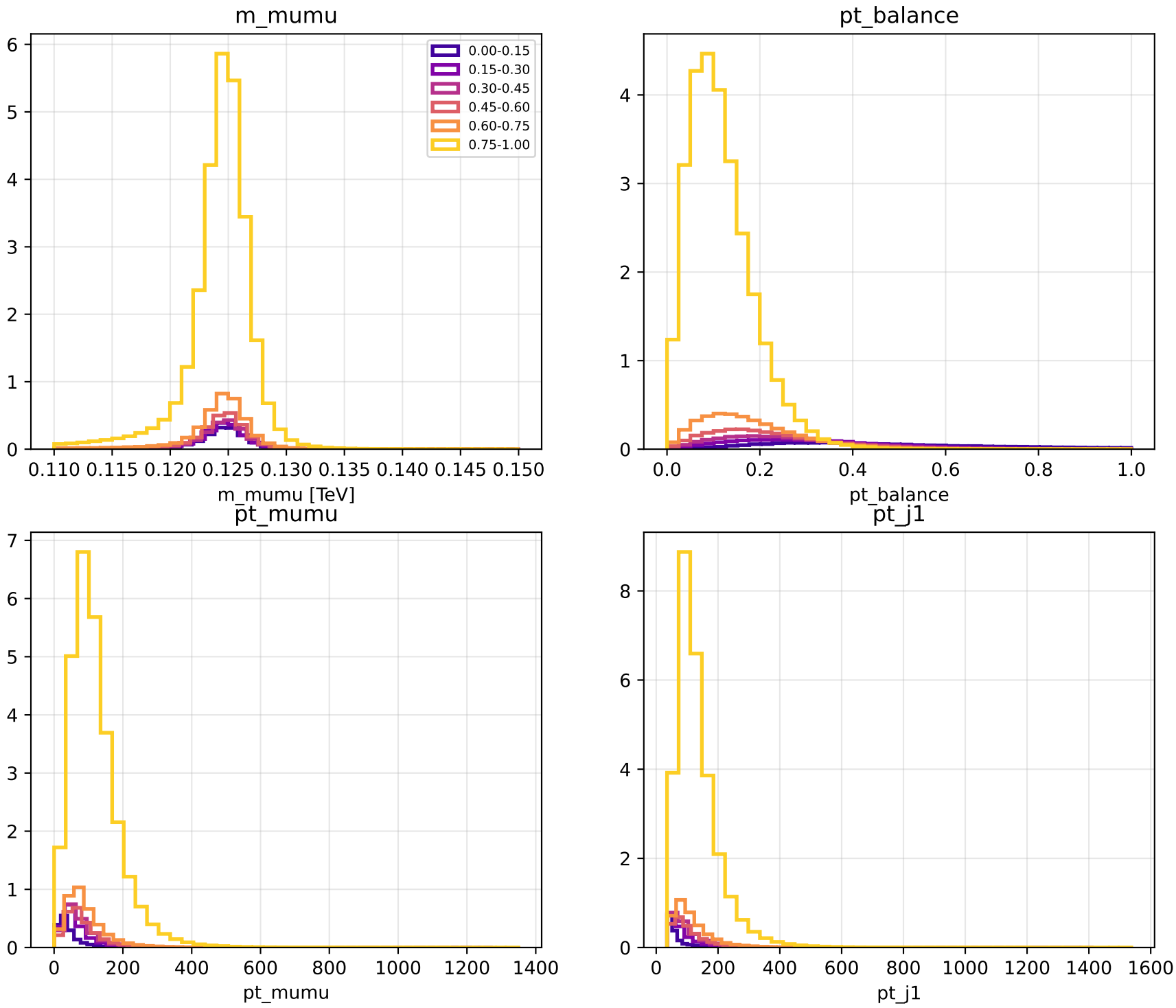
2024: Expected signal yield per BDT bin (110-150 GeV)

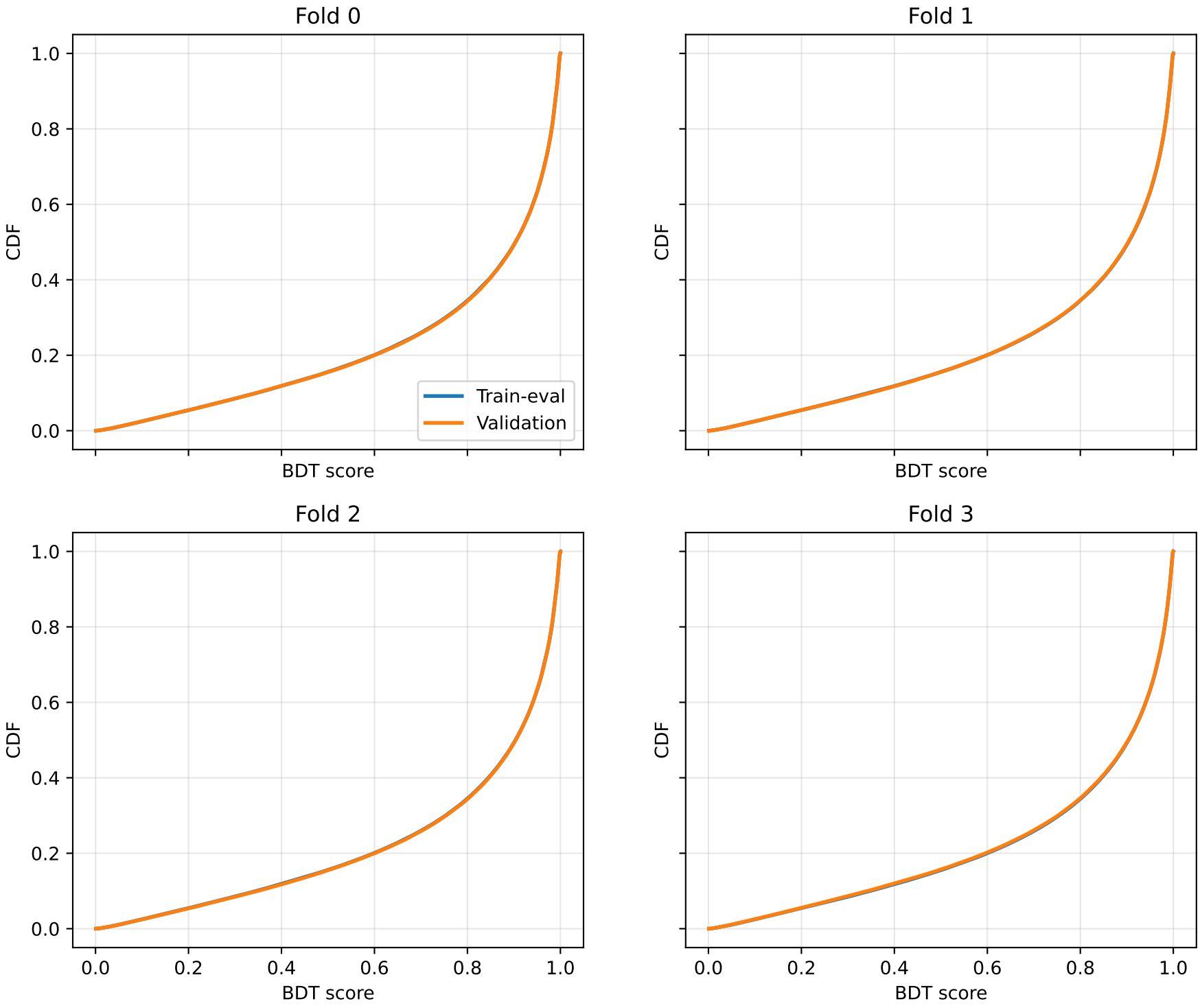
BDT bin	Yield
0.00-0.15	1.580
0.15-0.30	1.815
0.30-0.45	2.039
0.45-0.60	2.563
0.60-0.75	3.854
0.75-1.00	28.019


2024: Signal mass sculpting (TeV)
0.00 <= BDT < 0.15
0.15 <= BDT < 0.30
0.30 <= BDT < 0.45
0.45 <= BDT < 0.60
0.60 <= BDT < 0.75
0.75 <= BDT < 1.00
Weighted events
0
1
2
3
4
5
6
0.110
0.115
0.120
0.125
0.130
0.135
0.140
0.145
0.150
m_mumu [TeV]


2024: Background mass sculpting (TeV)
0.00 <= BDT < 0.15
0.15 <= BDT < 0.30
0.30 <= BDT < 0.45
0.45 <= BDT < 0.60
0.60 <= BDT < 0.75
0.75 <= BDT < 1.00
Weighted events
0
2000
4000
6000
8000
10000
0.110
0.115
0.120
0.125
0.130
0.135
0.140
0.145
0.150
m_mumu [TeV]

2024: Signal distributions by BDT bin




2024: CDF (train-eval vs validation)
Fold 0
Fold 1
Fold 2
Fold 3
CDF
BDT score
Train-eval
Validation
0.0
0.2
0.4
0.6
0.8
1.0

2024: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.19%	561778	0.001	0.20%	TRUE
1	0.16%	561778	0.001	0.20%	TRUE
2	0.19%	561777	0.001	0.20%	TRUE
3	0.25%	561777	0.001	0.20%	FALSE

2024: Top-10 feature importance (gain)
2024_featureimportance_background.txt

Feature	Gain %
pt_balance	33.48%
pt_mumu	11.43%
pt_j1	11.10%
z	9.63%
dphi_mumu_jj	7.52%
deta_jj	7.02%
pt_j2	6.56%
njets	3.13%
jet1_eta	2.27%
mu1_pt_over_m	1.77%